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Course: Calculus Beta

Assignment: MyLab Math - Uge 1

Find the derivative of the function $s = \frac{1}{3\pi} \sin(3t) - \frac{1}{7\pi} \cos(7t)$.

Use the rules of differentiation to find $\frac{ds}{dt}$.

$$s = \frac{1}{3\pi} \sin(3t) - \frac{1}{7\pi} \cos(7t)$$

$$\frac{ds}{dt} = \frac{1}{3\pi} \cdot \frac{d}{dt}(\sin(3t)) - \frac{1}{7\pi} \cdot \frac{d}{dt}(\cos(7t))$$

To find $\frac{d}{dt} \sin(3t)$, use the Chain Rule.

$$\frac{d}{dt}(\sin(3t)) = 3 \cos(3t)$$

To find $\frac{d}{dt}(\cos(7t))$, use the Chain Rule.

$$\frac{d}{dt}(\cos(7t)) = -7 \sin(7t)$$

Substitute the expressions for $\frac{d}{dt}(\sin(3t))$ and $\frac{d}{dt}(\cos(7t))$.

$$\frac{ds}{dt} = \frac{1}{3\pi} \cdot \frac{d}{dt}(\sin(3t)) - \frac{1}{7\pi} \cdot \frac{d}{dt}(\cos(7t))$$

$$= \frac{1}{3\pi} \cdot (3 \cos(3t)) - \frac{1}{7\pi} \cdot (-7 \sin(7t))$$

$$= \frac{1}{\pi} (\cos(3t) + \sin(7t))$$

Thus, $\frac{ds}{dt} = \frac{1}{\pi} (\cos(3t) + \sin(7t))$.